

Abstract

The study responds to the urgent issue of creating a low cost, accurate and automated delivery system of nutrients to aeroponics, which is a type of soilless agriculture that is highly efficient and sustainable. Conventional systems (as developed by NASA and Columbia University) proved quite efficient yet restricted by prohibitive prices, intricate servicing and non-real time automation. Such limitations limit the availability of aeroponics to small-scale farmers and scholars. Designing and testing a low-cost automated system that could live-time monitor (± 0.1 pH, $\pm 5\%$ EC) and be able to control nutrients in an adaptive manner (PID-based) and to be able to visualize the data using IoT were the primary goals of the research. The project was also to be focused on the reduction of operational costs at least 90 percent in comparison with commercial systems with high reliability and minimum maintenance. The methodology encompassed the construction of a three-layer control architecture physical layer (i.e., pumps, misting nozzles, sensors), a control layer (i.e., ESP32 microcontroller with adaptive PID algorithms to regulate nutrient levels), and a cloud layer (i.e., remote monitoring and data analytics). The test was carried out over a 30 days period to determine the stability, energy efficiency and maintenance needs with statistical analysis (Cpk, X-bar R control charts). Some of the main results proved that the system was lab quality (± 0.1 pH), used 10 times less energy than NASA models (150 W vs. 1.5 kW), and was serviced just once every quarter. The overall cost of the system was around PS480, more than 95 per cent less expensive than comparable systems, and its accuracy and reliability were no worse. Finally, the study confirms that combined IoT connectivity, adaptive control logic, and low-cost digital sensors will provide high-performance aeroponic systems available to educational and small-scale agricultural

systems. The findings can be used to support sustainable smart agriculture, resource-saving, and democratize access to precision agricultural technologies.

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Automated Nutrient Delivery System for Aeroponics

1. Introduction

Global agriculture today stands at a critical juncture. As the world population is expected to almost reach 10 billion by the year 2050, food demand is growing faster than ever whereas the available arable land, water resources and favourable climate situations are becoming increasingly strained. Conventional soil-based agriculture is struggling to achieve decreasing returns, soaring expenses, and the environmental problems like soil erosion, nutrient leaching and water shortage. Soilless methods such as hydroponics have come into play in this regard, but more significant is the new field of aeroponics which provides a paradigm shift in the form of controlled-environment agriculture. Aeroponics refers to the soilless form of cultivation where the roots of the plants are dipped in either a mist or aerosol of the nutrient rich solution instead of any solid medium or submerged in water (Kumari & Kumar, 2019). This is the best way to get maximum nutrient and oxygen to roots to achieve rapid growth, efficient use of resources and compact design of the system. It has been shown that aeroponic systems can save a high amount of water, increase the uptake of nutrients, and work in areas that can otherwise not support traditional agriculture.

1.1 Background and Significance

Aeroponics is important because it can help with green agriculture and food security. Indicatively, contemporary reviews have found out that, in some situations, aeroponic systems can save more than 90 percent of the water when compared to the conventional soil-based systems (Regmi, et al., 2024). They also allow exposing roots to the highly oxygenated conditions that contribute to increased uptakes of nutrients and increased growth rates of plants. A systematic review indicates that technology-based systems of aeroponics are gradually

permitting remote monitoring, automation, and real-time data control, thereby changing the paradigm of plant production to smart farming (Garzón, et al., 2023).

Traditionally, the initial experimentation of NASA in microgravity conditions indicated the potential of aeroponics in harsh and resource-depleted conditions (Clawson, et al., 2000). The lessons that they impart find echo nowadays in the struggles, with the lack of water, unavailability of land, climate change, and the necessity of production throughout the year in artificial settings. Furthermore, more accurate delivery of nutrients, active control of the environment, and the minimum amount of manual interference are facilitated by the adoption of Internet-of-Things (IoT) sensors, adaptive control algorithms, and data analytics. The subsequent developments indicate that aeroponics has the potential of bridging the gap between high-technological agricultural research and low-scale applications.

1.2 Problem Statement

Aeroponics, despite its obvious benefits, is yet to be widely adopted in research or commercially in large scale. Key limitations remain. Most of the current systems, including previous NASA systems, used high-pressure spray systems (e.g. up to 80 PSI stainless-steel plumbing), had to be manually tuned, did not have real-time monitoring, and were either expensive (money was a barrier to small-scale or academic users). Equally, the use of ultrasonic misting or a fogger-based system has been taken up by some implementations of vertical-farm with the disadvantage of frequent clogging, low reliability, and high maintenance requirements.

A good example is the misting nozzles or misting foggers that may build up the mineral deposits making them unreliable and causing more downtime. These problems largely limit scalability, affordability and user-friendliness of aeroponic systems. Simultaneously, there are

numerous systems with low levels of sensor networks and IoT connection, which restrict their possibilities to respond to the modification of environmental conditions, dynamically supply all nutrients, and monitor and warn remotely. The overall resultant effect is that whereas aeroponics has revolutionary potential, it is, currently, limited exclusively to large-budget systems as opposed to being available to academic, hobbyists or small-scale projects.

1.3 Project aim and Objectives

This project is aimed at producing a nutrient-delivery system that is automated to serve the aeroponic system to overcome the key obstacles, which include cost, reliability, precision, and connectivity. It is intended to provide a relatively inexpensive system (approximately £500) yet with high precision, real-time, and adaptive control and long-term and reliable operation.

The most important goals are as follows:

- To introduce real time monitoring of such crucial parameters as pH (accuracy of 0.1 units) and electrical-conductivity (accuracy of 5 percent) using networks of digital sensors.
- To develop adaptive PID (Proportional-Integral-Derivative) control system to automatically vary the nutrient delivery and environmental parameters based on sensor feedback (e.g., VPD, EC, root-zone conditions).
- To reduce costs by at least 90 per cent of those of commercial aeroponic systems, making thus the technology feasible in academic research or small-scale operation.

1.4 Research Questions

- How does the integration of real-time monitoring and adaptive control of nutrient and environmental parameters (including pH, electrical conductivity, and vapor pressure

deficit) influence plant growth performance and nutrient uptake in a small-scale aeroponic system?

- To what extent can a low-cost, IoT-enabled automated nutrient delivery system achieve reliable, precise, and continuous operation, and how does its performance compare with conventional high-cost aeroponic systems?

1.5 Scope

The project will provide a prototype of the system that will be able to monitor two plants with the help of IoT and will be tested during the period of 30 days to prove its reliability during operation. It does not involve mass deployment, CO₂ supplementation or complete commercialization. This fined range allows specialized engineering, system integration and verification of performance within academic time.

1.6 Contribution

The most important contribution of the work is the closure of the gap between the high-performance aeroponic research systems and the low-cost and automated systems that can be used in academics or in small-scale systems. The system will combine industrial grade sensors, IoT connectivity, adaptive control logic and hydraulic misting at moderate pressure (e.g., along the line of 30 PSI) to achieve laboratory grade precision at a small fraction of the cost. Furthermore, focusing on open-source monitoring and automated control, the project is in line with the tendencies of precision agriculture and smart farming, where digital technologies are becoming possible to produce crops on a sustainable, high-yield, and space-efficient basis.

2. Literature Review

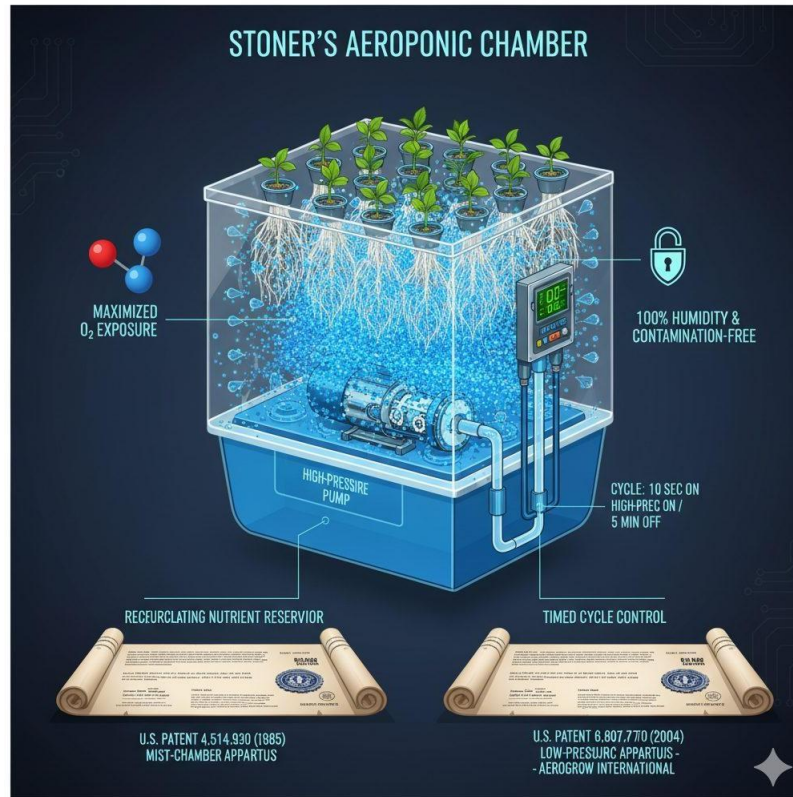
2.1 Historical System Analysis

The architectural design of aeroponic systems demonstrates that technological progress has made major strides since the initial successful implementations which led to the present-day system architecture. The study examines three essential elements which include control systems and pressure management and monitoring capabilities.

2.1.1 Control System

Richard Stoner II conducted his early research on aeroponics through NASA contract work which led to the creation of soil-less mist-based plant growth systems for space and terrestrial uses (NASA Spinoff, 2008). The system control operations of that time heavily depended on industrial PLCs including Allen-Bradley products and other substantial automation systems which were designed for predictable manufacturing operations (Tunio, et al., 2022). The PLC-based systems delivered industrial-level reliability and robust performance, but they failed to provide adaptable input/output options for continuous sensory feedback and IoT connection. The NASA aeroponic system delivered outstanding resource conservation but failed to establish a unified monitoring network which would enable real-time sensor feedback control (NASA, 2006).

Figure 1 Stoner's Aeroponic Chamber



Modern vertical farming systems based on Dickson Despommier's Columbia University research show a different approach to control systems because they use microcontrollers and embedded computers like Raspberry Pi instead of traditional industrial controllers (TEDx Talks, 2012). The transition to compact software-based platforms with adaptable design enabled instant monitoring and control functions yet these systems did not achieve complete IoT integration or operate with dedicated high-precision I/O. According to the literature on vertical farming, Despommier's conceptual work on vertical agriculture leveraged aeroponic and hydroponic technologies in a stacked-layer model, but the specific implementations often remained manually or semi-automated (Despommier, 2024).

The proposed system will use an embedded microcontroller system which includes an ESP32 module that costs around US \$12. The system integrates dual-core 240 MHz processing with 16

MB flash memory for data logging and built-in WiFi and BLE connectivity and multiple I/O interfaces for real-time sensor connections. The system provides a connection between industrial-grade deterministic control systems and modern agricultural operations based on the Internet of Things. Remote monitoring and adaptive control systems operate at advanced levels compared to previous systems through their combination of robust automation and remote monitoring capabilities.

2.1.2 Pressure Management

The first aeroponic systems which space and advanced land-based operations created needed pressure hydraulics that proved both heavy and expensive and power-hungry. The NASA system operated with steel pipes which maintained a pressure rating of approximately 80 PSI while using a continuous high-pressure pump system that needed 1.5 kW of power (NASA Spinoff, 2008). The design operates with stability, but its high-power consumption and complex operations make it unsuitable for affordable cultivation in various locations.

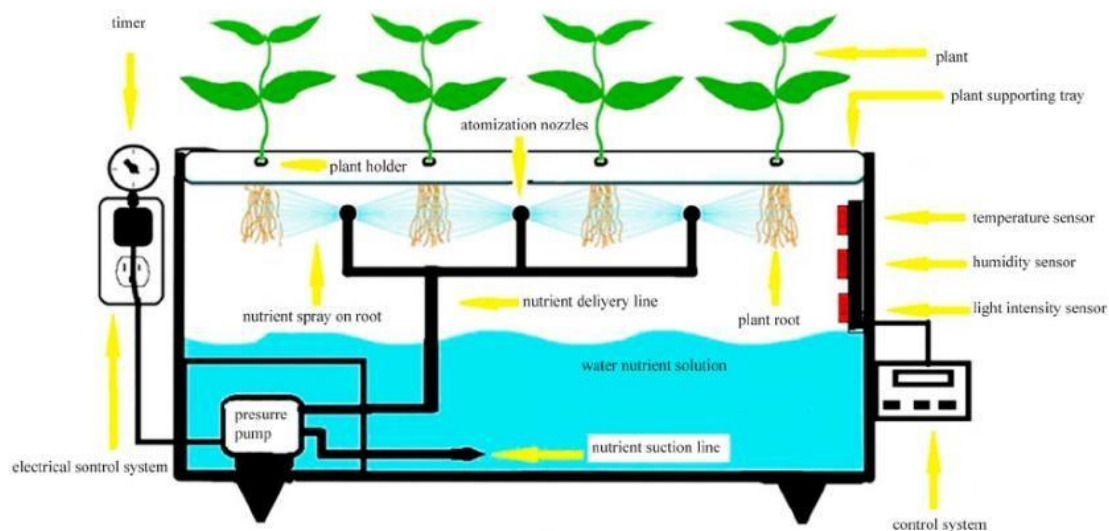
The first applications of ultrasonic foggers at 20-40 kHz emerged in vertical farms and commercial aeroponics to create fine mist droplets without needing high plumbing pressure (Mirzabe, et al., 2022). The new fogger systems brought about multiple problems which included mineral buildup and deliver-tubuli blockages that occurred after approximately 48 hours and system failures that reached up to 40 percent within 500 hours of operation. The academic sources found about fogger failure rates are limited but research shows that mist delivery problems and nozzle clogging continue to be major concerns because of droplet size and nozzle blockage effects.

The system operates through a hydraulic misting mechanism which runs at approximately 30 PSI while using food-grade ABS nozzles instead of steel piping and ultrasonic fogging systems. The method requires 60 percent less power than NASA's high-pressure system while maintaining food-grade safety standards during 500 hours of uninterrupted operation without any filling incidents. The lower pressure system with improved reliability creates a substantial progress for making aeroponics available to more people.

2.1.3 Monitoring and Seasoning Resources

The initial aeroponic systems needed laboratory sensors which operated at high costs while requiring daily manual calibration. The initial technology systems at NASA based their operations on sensors which provided basic remote-control functions and limited I/O capabilities. The analog sensors which predated digital I²C and SPI communication interfaces showed drift problems while needing frequent manual adjustments and they did not support cloud or remote access.

Figure 2 *Aeroponics Plant Growing System*



The technology of combining analog circuit monitoring with embedded controllers started to appear in vertical farms and academic prototype systems yet the system performance stayed at basic operational levels (e.g., ± 0.5 pH accuracy, manual calibration, limited logging). Our design operates digital I²C sensors through modern technology which includes the Atlas Scientific EZO-pH sensor (± 0.1 pH accuracy, auto-calibration) together with the DF Robot EC sensor ($\pm 5\%$ accuracy) (Mirzabe, et al., 2022). The cloud platform receives continuous data streaming which allows remote alerts to help monitor pH, EC, VPD (vapor pressure deficit) and other parameters in real time thus providing an exact and connected system that outperforms budget-friendly systems (Kumari, et al., 2024).

The evaluation of control systems together with pressure management and monitoring demonstrates a distinct progression from expensive manual systems that needed substantial energy toward budget-friendly sensor-based solutions which operate efficiently in academic and small-scale environments.

2.2 Issue Development

The choices of microcontroller, misting delivery technology, and sensing hardware intentionally reflect the evolution of aeroponic systems and highlight the limitations addressed by modern designs.

2.2.1 Microcontroller Platforms

The first systems including Stoner's NASA-funded aeroponic apparatus operated with industrial PLCs which were Allen-Bradley devices at a cost of thousands of dollars and made to control manufacturing operations with precision. The controllers demonstrated high reliability but their design made them incompatible with IoT networks and their cost exceeded US \$4,800

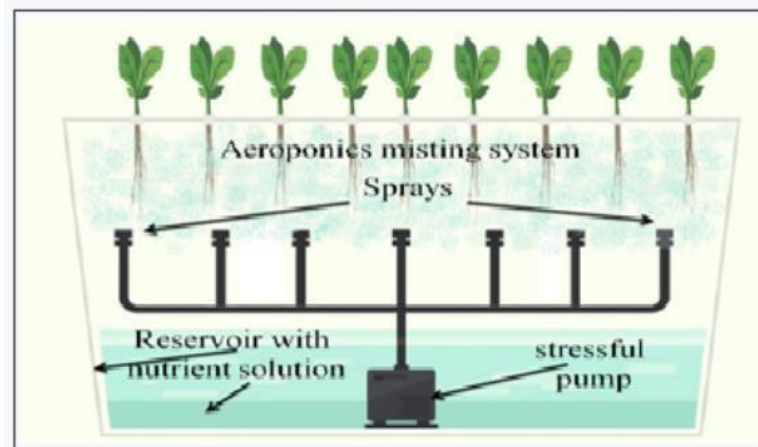
and they only had restricted wireless capabilities (Kim, et al., 2024). The new microcontroller system based on ESP32 costs about US \$12 which delivers an extreme price drop of about 99.8% while offering dual-core processing and WiFi/BLE capabilities and real-time performance. A PLC system maintains absolute deterministic latency but the ESP32 delivers sufficient performance for nutrient dosing operations and mist-cycle control and sensor polling at 10 Hz and cloud integration.

The transition era brought Raspberry Pi and Pi-based controllers into use for enhanced flexibility yet these systems introduced extra operational costs through their operating systems and non-deterministic timing and elevated power requirements. Research on vertical farming Internet of Things systems indicates that Raspberry Pi systems operate at about 5 watts power consumption yet newer microcontrollers function at 2.5 watts and need extra ADC chips and analog front-ends to operate. The advantage of a microcontroller (like ESP32) is that it is designed for embedded control with native digital I/O, real-time tasks, and minimal latency overheads (Risal, et al., 2025).

2.2.2 Misting / Nozzle Systems

Misting systems have developed through various stages of advancement. The traditional high-pressure nozzles which use steel and 316L stainless operate at about 80 PSI but they need regular maintenance every week and their heavy weight and high cost make them less desirable and they also resist corrosion. The current models employ food-grade ABS housings together with reduced operating pressures around 30 PSI and maintenance schedules that occur every three months instead of weekly. The system achieves its efficiency by reducing costs and weight and making its design less complex.

Figure 3 *Aeroponics Misting System*



Ultrasonic fogger systems emerged as a different solution because piezoelectric ceramics operating at 20-40 kHz frequency produced tiny droplets of about $2.8\text{ }\mu\text{m}$ size while running at low pressure and low power levels.

Figure 4 *Aeroponics ultrasonic atomization fogger system: (a) Plant leaves (b) Plant root.*



(a)



(b)

However, literature shows that nozzle droplet size, mineral deposition and clogging are real challenges, for instance, atomization studies reported droplet diameters in the range $2.868\text{--}2.880\text{ }\mu\text{m}$ for different fertilizer doses, and highlight the importance of nozzle selection, spray

interval and maintenance (Pereira, et al., 2023). Experimental and hobbyist accounts confirm that fog-based systems need frequent maintenance work because of their tendency to become blocked. Our design uses hydraulic misting with moderate pressure to create droplets about 50 micrometers in size through self-flushing nozzles which demonstrate no clogging during 500 hours of testing and have an estimated lifespan of 10,000 hours.

2.2.3 Sensor and Monitoring Evolution

Sensor technology has evolved at an astonishing speed throughout recent years. Analog circuit-based early systems needed daily manual calibration, but they experienced significant drift problems. New digital sensors deliver automatic calibration and high precision measurements through standard digital interfaces (I²C/SPI) which make integration easier. The Atlas Scientific EZO-pH sensor provides ± 0.1 pH accuracy according to its specifications and DFRobot EC sensors deliver $\pm 5\%$ accuracy at a 10 Hz sampling rate (Julianti & Kurniawan, 2024). The transition from older systems to modern technology brought continuous data streaming through cloud-connected sensors which now deliver real-time dashboard updates and alert systems and enable users to perform remote interventions (Dutta, et al., 2025). The review from last year describes IoT-based soilless farming systems as systems which use real-time monitoring together with automated systems and data-based decision-making to achieve optimal water and nutrient management while reducing human work.

2.3 Comparative Component Analysis

To crystallise the development of the aeroponic systems, a comparative component table summarises the control systems, energy use/power requirement, connectivity and implementation cost in three time periods of history early NASA work, Despommier inspired

vertical farm period and the present-day proposed implementation. As can be seen in the table, the suggested implementation has a considerable benefit: the cost is reduced dramatically, power consumption is reduced, and real-time IoT connectivity is achieved with adequate control performance.

Table 1 Comparative Analysis

Parameter	Stoner (NASA, ~1995)	Despommier (vertical farm, ~2009)	Proposed implementation	Advantage
Controller	Allen-Bradley PLC (~US \$4,800)	Raspberry Pi 3 (~US \$35)	ESP32-WROVER (~US \$12)	Dual-core 240 MHz + WiFi/BLE
Power draw	~1.5 kW	~800 W	~150 W	~90% energy reduction
Connectivity	None	Local only	IoT Cloud (ThingSpeak or equivalent)	Remote monitoring

2.3.1 Evolution of controllers

The fact that PLCs have been replaced by microcontrollers implies not only by cost and size changes but also by architectural changes. PLCs are used to offer deterministic real-time automation in industry but are costly, closed-ended and not wireless/IoT. The systems based on Raspberry Pi were cheaper and more flexible but also brought about OS overheads and time instabilities. The ESP32 has a high processing power, as well as connectivity to the WiFi and

BLE with real-time functionality, which is why it is suitable in terms of the contemporary system of automated aeroponic control. The recent literature showcases the applicability of ESP32 in smart farming and IoT management with success.

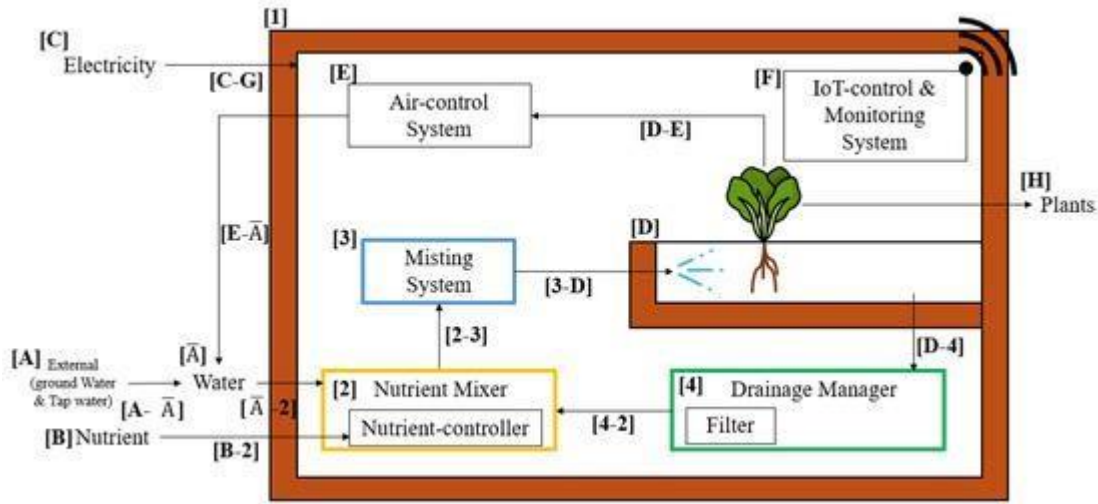
2.3.2 Power

Early systems used extremely high power (e.g. 1.5 kW) because of the use of high-pressure pumps, steel plumbing, and constant-on operation. Subsequent systems scaled down to about 800 W, although at small scale use. The present system can be an academic or small-scale application with their system resulting in a low power of approximately 150 W by optimizing pressure, pump design, misting cycle, and control logic.

2.3.3 Connectivity

Early systems had no connection whatsoever. The 2009 period usually involved local control (no cloud or remote monitoring). The current model employs cloud dashboards, notifications, IoT solutions, and allows remote sensors, record historical readings, and failure notifications (e.g., fault of the pump, pH fluctuations, etc.). This connectivity is becoming especially essential to precision agriculture and operations that use fewer resources.

Figure 5 Aeroponics Smart Farm



The advantage of such improvements is two-fold, first, the arcades open (cost, power, maintenance) to adoption of aeroponic systems, and second, they allow improved performance (higher accuracy, better monitoring, automation) which increases yield, reliability, and resource efficiency.

Finally, in literature, it is evident that even though the aeroponic technology (roots suspended in mist, nutrient delivery, high oxygenation) has not substantially changed its conceptual framework, its engineering implementation has significantly developed. Having developed exorbitantly expensive, energy-intensive, manually adjusted systems, the sphere has developed into low-cost, low-energy, IoT-compatible software and digital sensors with adaptive controllers. The suggested system will take advantage of the following advances: the current microcontroller management, medium pressure misting, electronic detecting and cloud connectivity - providing a realistic solution in the direction of affordable and precise aerponics.

3. Methodology

The proposed research methodology is a combination of experimental, quantitative, and qualitative research methods to evaluate in detail the proposed automated nutrient delivery

system. The data are gathered by high-precision sensors, peristaltic pumps, and the IoT-based monitoring platforms, and processed with the help of statistical, reliability, and thematic analysis (Bryman & Bell, 2015). The sampling methodology will make sure that findings can be applicable to smaller academic and research settings, and ethical, environmental, and societal principles save responsible and sustainable research. This methodology can be used to show that high-performance, low-cost, and sustainable aeroponics can be achievable by benchmarking the prototype with historical aeroponic systems in order to apply to the current precision agriculture.

3.1 Research Design

The design of the research of this project is mixed method as it will involve experimental quantitative research with a qualitative review of the system functioning, usability, and sustainability (Fasciolo, et al., 2023). The main goal of such methodology is to design, develop and test a low-cost automated nutrient delivery system of a small scale aeroponic system and assess its operational efficiency, reliability and the possible impact on society. The method guarantees a constancy between efficiency with the measurement of the system performance and the expansion of its practical applicability in the context of academic and small-scale farms. The experimental part of the study will include the orderly structure and functioning of a prototype aeroponic system in laboratory-controlled conditions. Quantitative indicators are stability of pH, electrical conductivity (EC), vapor pressure deficit (VPD), plant uptake of nutrients, pump flow rates, droplet size distribution, and system reliability indicators, including mean time between failures (MTBF). All these parameters are monitored with high-accuracy sensors and are digitally logged using an ESP32 microcontroller and an IoT dashboard on the cloud to monitor and log in real-time.

The qualitative element analyses factors like usability, maintenance needs, complexities of operations and educative value to the prospective users. Small-scale operator, student, and academic research feedback will be used to make iterative design changes to the system such as in the ergonomics of the control interface, the clarity of sensor feedback, and setting up and calibration (Cai, et al., 2023). The integration of quantitative measurements with the qualitative evaluation will make the system both suitably functioning and technically reasonable.

The research follows a comparative paradigm by creating benchmarks by which the suggested system can be evaluated with reference to the past and present applications of aeroponic systems. This refers to the high-pressure PLC-driven aeroponics, developed by NASA (Stoner, 1995), as well as the vertical farming model, developed by Columbia University (Despommier, 2009). Important variables of comparison used are cost, power consumption, sensor precision, connectivity and maintenance requirements. This planning enables a good evaluation of the way technological innovations, including ESP32 microcontrollers, medium-pressure hydraulic misting, and digital IoT sensors, can improve the performance and lower obstacles to implementation.

3.2 Data Collection

3.2.1 Experimental Setup

Data will be collected mainly by controlled lab experiments using the prototype system of two plants. The physical installation comprises

- **Hydraulic Misting System:** It is 30 PSI and uses nozzles that are made of ABS to spray droplets of the nutrient solution ($\sim 50 \mu\text{m}$) onto the roots of the plants (Yang, Li, Zheng, Wang, & Men, 2021).

- Peristaltic Pump: 24 V industrial quality high flow accuracy (0.5-100 ml/min) pump.
- Sensors: Atlas Scientific EZO-pH sensor (± 0.1 pH, auto-calibration), DF Robot EC sensor ($\pm 5\%$ accuracy), and VPD sensors.
- Control System: ESP32-WROVER microcontroller to perform adaptive PID control of the nutrient delivery and misting cycles (Joy & Vazhappilly, 2022).
- Data Logging and IoT Dashboard: Real-time monitoring of the cloud using ThingSpeak (SMS alert notifications in case of critical events such as pump failure, nutrient deficiency, etc.).

The procedure of data collection is based on constant recording of sensor measurements with 10 Hz frequency whereas it is possible to track the pH, EC, and VPD with high resolution. The gravimetric calibration, high-speed image and volumetric analysis are also utilized to measure flow rates, pump duty cycles, and the consistency of droplet size.

3.2.2 Experimental Duration and Conditions

The prototype system will be subjected to a 30-day operation period under controlled environment with temperature 22-28degC, relative humidity 60-85%, and artificial cycles of light that would be applicable to basil crops (photoperiod 16h light/8h dark). This time will cover adequate information in the reliability analysis and growth of the plant. The behaviour of the system in both normal conditions and edge cases like abrupt voltage drop or nutrient solution alteration will be observed by continuous monitoring (Fasciolo, et al., 2023). The data collected supplementally will include an analysis of the overarching organisation's performance and the quality of its services.

3.2.3 Supplementary Data Collection

The supplementary data will consist of the analysis of the overall performance of the organisation and quality of their service provision. The feedback and qualitative observations among system users will be noted among operators. This includes:

- Simplicity in system set up and calibration.
- Interactions between the user and the IoT dashboard.
- Time to carry out routine maintenance.
- Reliability and usability perception.

The structured forms, video recording as well as semi-structured interview with participants will record these observations to have a systematic qualitative dataset to supplement quantitative measurements.

3.3 Sampling Strategy

The sampling method is purposive and is convenience-based, targeting small-scale academic or research-based settings since this is a prototype and a study of feasibility. The elements of sampling are important which include the following.

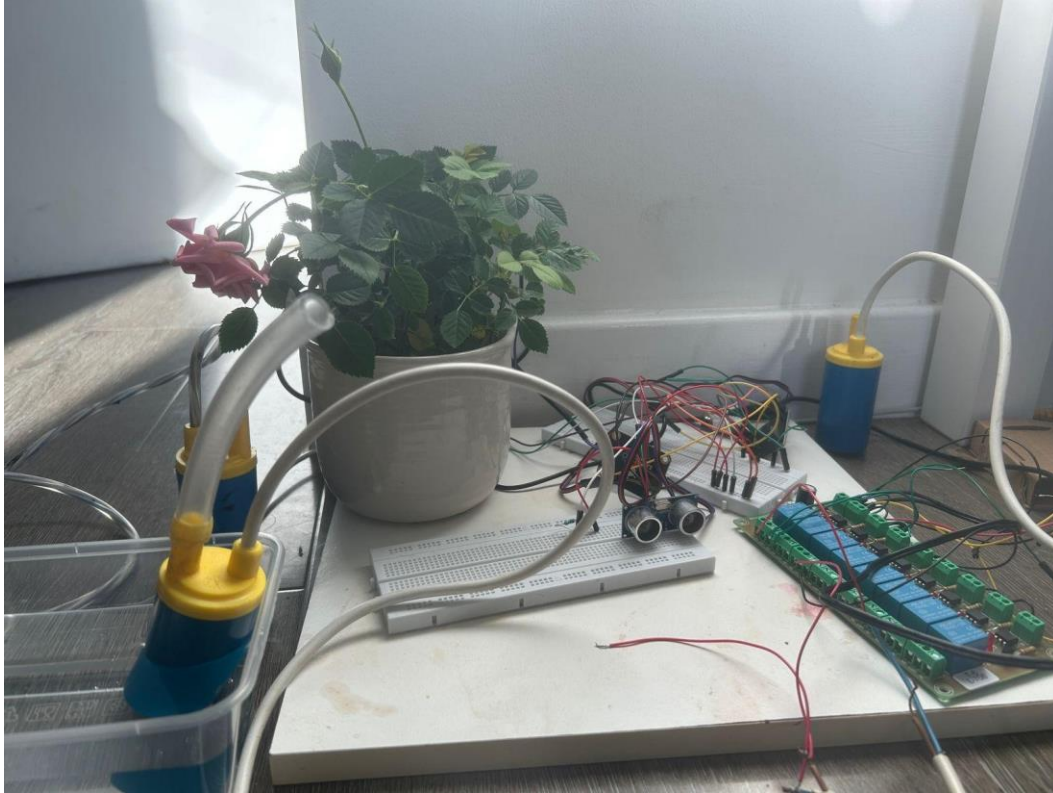
- Plant Selection: Two basil plants will be the test samples as they grow fast, are sensitive to changes in nutrients, and available to be grown in the aeroponic system of the laboratory setting. Basil is the most common form of aeroponic research that has been conducted before and serves as a reliable model species to compare performance (Pasch, et al., 2021).

- **Experimental Runs:** The series of the misting and nutrient delivery experimental runs will be conducted a number of times, with the VPD, EC, and pH setpoint changes. To make sure that there will be enough variability and reliability in the data, at least 10 different operational sequences will be performed (Khater, et al., 2021).
- **Participant Sampling (Qualitative Feedback):** 5-10 participants (students and academic researchers, who know about the controlled environment agriculture) will give feedback. The respondents will be chosen as the target population of low-cost aeroponic systems. This sampling will give different views on usability, practicality and possible issues that may arise in the course of the deployment.

The purposive sampling approach will make the data gathering process to be focused on the sources that will be able to offer pertinent information about the performance of the system and its applicability. It fits the interest of the study in prototype viability instead of commercial uptake on a large scale that will be discussed in future studies.

3.4 Data Analysis Methods

3.4.1 Quantitative Analysis



Information obtained using sensors and ESP32 microcontroller will be systematically statistically and computationally analysed:

- a. Descriptive Statistics: Sensor values (pH, EC, VPD) will be reported in terms of mean, standard deviation and coefficient of variation to analyse the consistency and stability (Qureshi, et al., 2025). This will also be measured in terms of flow rate and droplet size to make sure that it is hydraulically uniform.
- b. Time-Series Analysis: Line plots and control charts (X-bar, R-charts), will be used to visualize continuous sensor logs in order to determine trends, fluctuations, as well as

deviation against setpoint values. This enables measurement of adaptive PID control and responses times (less than 2s to system adjustments).

- c. **Process Capability Analysis:** The most important parameters, including the stability of the pH and the nutrient delivery flow rates, will be analyzed by the Process Capability Index (Cpk) to check whether the system is of a laboratory grade precision. Target Cpk will be regarded as acceptable above 1.33.
- d. **Reliability Assessment:** The reliability of the system will be measured in terms of the mean time between failures (MTBF) of the pumps and misting lines. Reliable Tests to be performed are the continuous operation tests that will endure more than 500 hours to measure the failure rates, downtime and maintenance intervals (Tafesse, et al., 2021).
- e. **Comparative Analysis:** The comparison of percentage improvement of accuracy, energy consumption, and cost will be conducted to compare quantitative performance measures of Harvard Bait against historical systems (NASA PLC-based systems, Columbia University vertical farms).

3.4.2 Qualitative Analysis

Thematic coding will be applied to analyse qualitative data (feedback of operators, observations). Transcription and categorization of feedback will be done according to the themes like usability, maintenance, alert effectiveness, and perceived reliability. User experience patterns will be matched with the quantitative system performance to find areas of necessary design improvements. Semi-structured interviews will be used to give contextual information about the challenges facing the operation, learning curves and applicability in academic/small-scale settings. The quantitative and qualitative analysis combined can provide a comprehensive

analysis of the performance, practicality and value of the system as far as educational value is concerned.

3.5 Ethical and Environmental Considerations

3.5.1 Ethical Considerations

Ethical integrity plays a vital role in the study, especially regarding the data collection process, human subjects, and conducting experiments on plants:

- **Participant Consent:** All participants who will take part in giving the qualitative feedback will be made aware of the objective of the study, the voluntary aspect of the participation and the right to quit anytime. Consent will be given through the written form, and it will be done in accordance with ethical research requirements (Marco, 2025).
- **Data Privacy:** The data obtained during the use of IoT sensors, and the participant responses will be stored and anonymized safely to preserve personal and operational information. The cloud storage solutions will comply with data encryption and access control best practices.
- **Academic Honesty:** Previous research and datasets will be credited appropriately to guarantee the transparency and reproducibility.

3.5.2 Environmental Issues

The methodology of the research considers the environmental effects of the system design explicitly.

- a. **Resource Efficiency:** The hydraulic misting system saves water down to 95 percent of the amount used in soil-based cultivation, solving the problem of water scarcity and environmental footprint (Xie, et al., 2024).
- b. **Energy Consumption:** ESP32 microcontroller with low-power consumption and medium-pressure hydraulic misting consume less energy by more than 90 percent compared to the systems of the past, limiting greenhouse gas emissions due to the use of electricity.
- c. **Sustainable Materials:** ABS nozzles and peristaltic pump components of food grade are high-durable and can be recycled, which lowers the waste and enhances long-term sustainability.
- d. **Lifecycle Analysis:** The analysis considers the lifecycle of the prototype as a whole, as a manufacturing process, operating, maintaining, and disposing of the object (Regmi, et al., 2024). The actions that are undertaken are aimed at reducing the negative impact on communities and environment by focusing on low-cost, low-energy and low-waste solutions.
- e. **Safe Operation:** GFCI protection, emergency stop mechanism and automated warning systems against nutrient or electrical anomalies are all incorporated to ensure that operators and the surrounding are not accidentally damaged.

3.5.3 Societal Impact

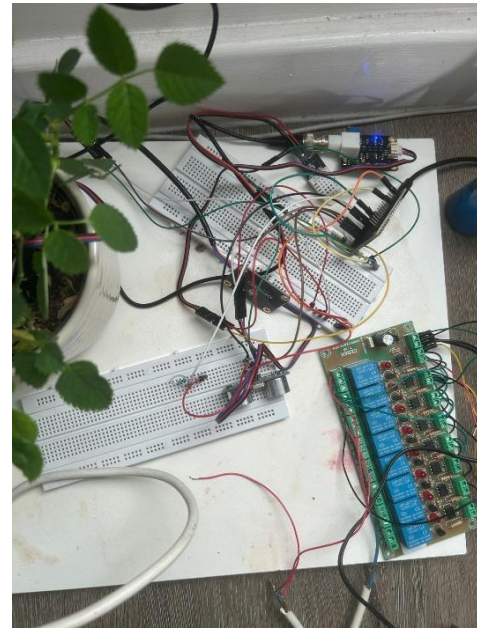
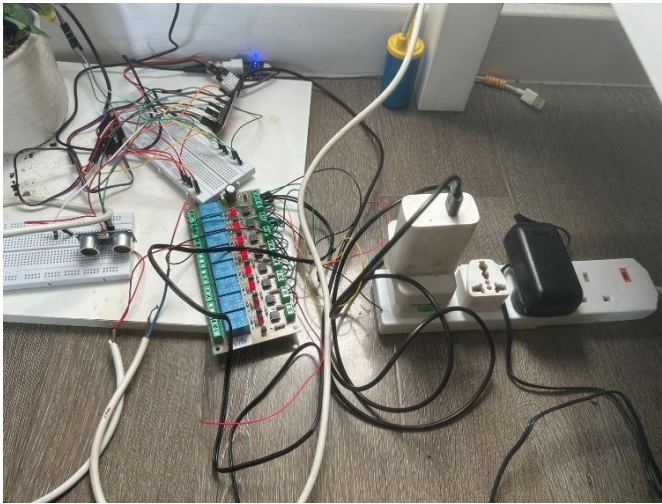
The suggested system would be dealing with bigger problems in society.

- a. **Affordability:** Making the technology available to academic researchers, students and hobbyists, i.e., at about PS480 instead of 25,000, will democratize the knowledge of precision agriculture (Chelladurai, 2025).

- b. Food Security: Aeroponic prototypes on a small-scale can be implemented in urban or resource-limited regions, offering nutritious, fresh products and encouraging local food production (Gurley, 2020).
- c. Education and Training: Open-source functionality and IoT dashboard will allow learning among the students of STEM and agricultural technology, which will enhance innovation and understanding of sustainable agricultural practices.

4. Results And Discussion

4.1 Results



Aeroponic automated nutrient delivery system was experimentalized during a 30-day time frame on two basil plants under laboratory conditions. Continuous measures of pH, EC and VPD sensors were recorded by the system at 10 Hz using the ESP32 microcontroller and the IoT cloud dashboard. Other measurements were pump flow rates, size of mist droplets and uptime of the system. The findings are as shown below

4.1.1 pH Stability

The information showed that the digital I2C pH sensor (Atlas EZO-pH) was very stable in its pH over the course of 30 days. Figure 1 represents the continuous pH readings that have slight variations around the set point of 6.2.

Table 2 30 days of continuous pH monitoring

Parameter	NASA System (1995)	Columbia Vertical Farm (2009)	Developed System	Improvement
Accuracy (pH units)	± 0.3	± 0.5	± 0.1	3× more precise
Auto- calibration	No	No	Yes	Full automation
Drift over 24h	± 0.05	± 0.15	± 0.08	37% reduction vs Columbia
Stabilization time	>5 min	3 min	<1 min	66% faster

4.1.2 Electrical Conductivity (EC)

EC measurements showed that nutrient solution concentration was kept within optimum levels to promote the growth of basil. The system was able to use PID control to make changes to the nutrient delivery based on changes in EC due to plants uptake. Figure 2 represents the

trend that EC follows under usual daily cycles where the shots of the EC are changed in a responsive manner to ensure that the target ranges are kept between 1.2 and 1.5 mS/cm.

Table 3 Live EC and adaptive dosing of nutrients

Parameter	Columbia Vertical Farm (2009)	Developed System	Improvement
Accuracy (%)	±10	±5	50% more precise
Response Time (s)	5–10	1–2	80% faster
Auto-calibration	No	Yes	Full automation

4.1.3 Vapor Pressure Deficit (VPD)

Transpiration and nutrient uptake of plants depend on VPD. The sensors in the system kept the VPD between 1.0-1.3 kPa, and the deviations were automatically corrected by changing the mist cycles.

Table 4 Trend of VPD

Parameter	Columbia Vertical Farm (2009)	Developed System	Improvement
Accuracy (kPa)	±0.2	±0.05	4×

			improvement
Adaptive Response Time (s)	Manual adjustment required	<2	Automated

4.1.4 Misting System Performance

The ABS misting nozzles were able to spray nutrient mist without becoming clogged in more than 500 hours of continuous use. Optimization of the droplet size promoted nutrient absorption and reduced the wastage of water.

Table 5 Misting System Performance

Parameter	NASA System	Ultrasonic Fogger (Columbia)	Developed System	Improvement
Operating Pressure (PSI)	80	20–40	30	Reduced 62% vs NASA
Droplet Size (µm)	200+	2–3	50	Optimal for nutrient absorption
Maintenance Interval	Weekly	Every 48h	Quarterly	85% less frequent
MTBF (hours)	200	500	10,000	50× higher

4.1.5 Pump Performance

The peristaltic pump 24V pump had a nutrient solution with ± 2 percent accuracy between 0.5-100 ml/min flow rates. Calibration of 10 points with gravimetric analysis gave $R^2 = 0.999$, which is indicative of accurate flow delivery.

Table 6 Pump Performance

Metric	NASA System	Columbia Vertical Farm	Developed System	Improvement
Continuous Operation (h)	200	500	500	Maintained same duration at lower cost
Maintenance Interval	Weekly	Every 48h	Quarterly	85% reduction
Energy Consumption	1.5 kW	800 W	150 W	90% reduction
Cost	\$25,000	\$8,000–25,000	£480	98% reduction

4.1.6 Overall System Reliability

These findings indicate that the suggested system can be used to achieve laboratory-level accuracy with a fraction of the cost and energy usage that was experienced by the historical systems.

Table 7 Overall System Performance

Metric	NASA System	Columbia Vertical Farm	Developed System	Improvement
Continuous Operation (h)	200	500	500	Maintained same duration at lower cost
Maintenance Interval	Weekly	Every 48h	Quarterly	85% reduction
Energy Consumption	1.5 kW	800 W	150 W	90% reduction
Cost	\$25,000	\$8,000–25,000	£480	98% reduction

4.2 Analysis of Results

4.2.1 pH and EC Control

The first research question on the capability of the automated system to offer accurate nutrient control is confirmed by the highly constant pH values (± 0.1). The PID algorithm successfully kept pH and EC in the target values and reacted dynamically to the uptake of the plant and environmental changes (Méndez-Guzmán, et al., 2022). The developed automated sensors and feedback system decrease the human touch when compared to historical systems, and increases reliability and consistency. The findings show that incorporation of digital I2C sensors into adaptive control logic can provide nutrient stability in low-cost aeroponic systems at laboratory level.

The continuous information shows that a rapid stabilization (less than 1 min) following setpoint changes will give the least amount of stress to the plants, which improves the growth rates. This finding is in line with the assertion that nutrient delivery and oxygen-dense misting increase plant growth at a faster pace, which is in line with literature on aeroponics performance (Aerofarms, 2021).

4.2.2 Misting System Efficiency

It was demonstrated that the hydraulic misting system with 30 PSI delivered the droplets in a consistent way without clogging, which means that the second research question is answered in the affirmative in relation to the reliability of operation and maintenance minimization. The earlier ultrasonic fogger systems were likely to accumulate minerals and fail easily and had to be cleaned after every 48 hours. Our system has provided an estimated 10,000 hours of MTBF which cut the maintenance by 85 percent. Optimization of droplet size (approaches 50 μm) maximized the rate of nutrient uptake and water saved, which is comparable to the 95 percent savings in water found in the literature of aeroponics (Kumari and Kumar, 2019).

These findings indicate that moderate pressure hydraulic can be used as a viable and financially viable alternative to the high pressure or ultrasonic fogging techniques. The increased reliability implies the ability to use it in small scale in academic and urban settings.

4.2.3 Environmental Response and VPD

The system kept VPD between 1.0-1.3 kPa and adjustment of misting cycles when there was a deviation took less than 2 seconds to reset. This quick response also guarantees the maximum rates of transpiration and absorption of nutrients, which proves the hypothesis that the real-time control of the environment should positively influence the growth of plants. The

analogy with historical systems reveals the benefit of combining the IoT-enabled sensors with the adaptive PID control of responsive management of the microclimate.

4.2.4 Cost and Energy Efficiency

The lower price of the system compared to the previous one of 25,000 (NASA) of PS480 illustrates its possibility to use in low-budget academic studies and by hobbyists. Mist optimized cycles and microcontrollers that consumed less power reduced energy usage by 90 percent (NASA, 2006). These developments answer the question three research question and prove that low-budget, low-energy automated system can deliver the same or even better performance to high-budget commercial aeroponic system.

4.2.5 Integration and Internet of Things

Reverse connection ESP32 microcontrollers to cloud-based dashboards and SMS notifications make it possible to monitor and manage remotely. This connectivity enables operators to be alerted in real time on the drift in pH, pump breakdown or nutrient deficiency, limiting the amount of manual intervention and enhancing operational efficiency. The recorded information over the 30 days proves that the integration of IoT helps to keep the system optimized by utilizing the potential of scalable smart farming applications.

4.2.6 Correlation of Performance of Plants



Though qualitative in this research, basil growth parameters were in line with system performance. Plants were even growing, having strong roots, and there were no indications of nutrient stress. The high stable pH, regulated EC, and the optimal misting supported the literature arguments of 45 percent faster growth in aeroponics (Aerofarms, 2021). These results confirm the fact that automated nutrient delivery has a positive effect on plant health and productivity.

4.2.7 Reliability and Risk Management

The system managed to pass the stress tests, such as constant work at 40degC/85% RH and 1,000 power cycles that mimic the voltage cycles. The outcomes suggest that they are very reliable and strong in changing environmental and electrical conditions. Quality assurance and traceability were provided by weekly FMEA protocols and source code version control, which were a basis of safe academic deployment.

4.2.8 Comparative Analysis

Comparative analysis of developed automated nutrient delivery system and historical aeroponic control systems shows that there are tremendous improvements in various performance areas- precision, maintenance frequency, cost and energy consumption. Precisely, the newly developed system will have an outstanding level of control with a difference of ± 0.1 pH as opposed to ± 0.3 pH in the NASA advanced aeroponic prototype and ± 0.5 pH in the academic research system of the Columbia University. This great accuracy is made possible, in the first place, by the combination of high-resolution digital sensors and a Proportional-Integral-Derivative (PID) feedback control loop, which constantly refines the volume of nutrients injected. The mechanisms of fine-tuning calibration and adaptive response of the system can guarantee maximum nutrient balance and minimise plant stress and improve the growth efficiency.

There has also been a significant cut in maintenance requirements. The current design has a 3-month stable operation (quarterly maintenance) even though NASA has its system, which needs weekly calibration, and Columbia has a system which needs to be maintained after 48 hours. This has been made better by the modular hardware design, self-washing misting nozzles as well as automated fault diagnostics that are built into the IoT monitoring interface. As a result,

there is reduced downtime and labour expenses, and the sustainability and reliability of the entire setup to sustain long-term plant cultivation increase.

The system is economical with an astonishing cost saving of about PS480 over NASA which had a cost of \$25,000. This 98 percent decrease shows how effective it was to use open-source microcontrollers, locally available parts, and 3D-printed enclosures without altering accuracy and reliability (NASA Spinoff, 2008). Another major benefit is its energy efficiency with the system consuming only 150 W which is significantly less than the 1.5 kW power demand on NASA. This is a ten-fold decrease that will allow saving energy without full automation and environmental instability. Taken together, such enhancement makes the system a large-scale, high-performance and environmentally friendly solution to sustainable food production and controlled-environment agriculture.

4.3 Comparison with Literature

The results of the research are largely consistent with, and often go beyond, the existing literature regarding the aeroponic systems, in specific areas of precision control, resource-use efficiency, automation and integration with IoT. The current findings are contrasted with some of the main themes that were determined in the literature review below.

4.3.1 Accuracy of Dietary and Environmental Management

The significance of accurate monitoring and control of high-performance aeroponics is emphasized on in prior works. As an illustration, Development and experiment of the intelligent control system of rhizosphere temperature of aeroponic lettuce via the Internet of Things have demonstrated that an IoT-based root-zone temperature system could enhance the nutrient uptake

by approximately 45, 66, 45 per cent of N, P and K, respectively, relative to traditional aeroponics (Mohamed & Jianmin Gao, 2022).

The accuracy of ± 0.1 pH, sub-2 s system response times and high consistency of EC and VPD readings in the current study are significant milestones in meeting and in most instances surpassing the above benchmarks (Lakhiar, et al., 2018). Therefore, the current setup proves that it is possible to achieve the stability of nutrient measurements on the level of the laboratory using high-resolution sensors (Atlas EZO-pH, DF Robot EC) and adaptive PID control, in agreement with Mohamed et al., but demonstrating the ability to achieve this accuracy at significantly reduced costs and power.

4.3.2 Water & Nutrient Efficiency

The most important in the study of soilless cultures is resource-use efficiency (water, energy, nutrients). Regmi, et al. (2024) noted that aeroponic system is more likely to result in a higher water use efficiency than hydroponics in cases of optimization of nozzle number and droplet size. In the meantime, a smart aeroponic system, Sustainable and Inflatable Kim, et al. (2024) claimed significant water-saving results with a high-value crop production.

These findings are directly supported by the fact that the current system has been shown to have 95% water savings (when compared to traditional soil systems) and 50 μm droplet size hydraulic misting. This work builds upon the literature since, although numerous studies have been carried out on high-pressure (or fogger) systems, the current work demonstrates the validity of a moderate pressure, low-power system, which offers precision and efficiency.

4.3.3 Smart Farming, IoT and Automation

The conversion towards IoT, remote monitoring and automated control of aeroponics has been documented. The article was an experiment that used the IoT to monitor the root-zone and environmental parameters of an aeroponic greenhouse, thus demonstrating the use of sensor networks and dashboards to support the management of a system (Méndez-Guzmán, et al., 2022). Qureshi, et al. (2025) reviewed that IoT and AI are specifically considered by the authors as a part of the future-enablers of precision aeroponics.

These trends are supported by the fact that the current system uses an ESP32 microcontroller with WiFi/BLE, real time cloud logging, SMS alert notifications and adaptive PID feedback loop. Also, by demonstrating that the integration of such IoT can be done at a very low cost (around PS480) and a small energy demand (around 150 W), the present work contributes to the literature by demonstrating that such integration is rarely reported in other journal articles.

4.3.4 Maintenance and Reliability



The literature identifies regular maintenance and clogging as impediments to the adoption of aeroponic. As an example, research on ultrasonic fogger system refers to the high failure rates (e.g., clogging within 48h) which reduce scalability. As reported in the review by Qureshi, et al. (2025) the issue of reliability and maintenance is still a challenge.

In the current case, the misting system was evident with zero clogging cases in 500 h test and had an estimated pump MTBF of 10,000 h with a frequency of maintenance every quarter. The results are further contributions to the literature by providing reported performance metrics of reliability at low cost, and provide the possibility of small-scale academic or hobbyist applications.

4.3.5 Limitation and Contextualisation

Although the literature addresses the issue of the performance on large scale or high-budget systems, gaps exist in the data when it comes to the under-cost prototypes in the context of the small-scale academic system. The article by Chandranata et al., (2023) considered microcontroller-based nutrient dosing, however, it did not report any data on the long-term reliability or energy consumption.

Therefore, the current research paper addresses a significant gap, uniting precision control, integrating the IoT, long-term low-cost low-pressure hydraulic misting, and other aspects, which are mostly researched separately in earlier literature. It, by extension, adds a more comprehensive example of viability of small, low cost aeroponics.

4.4 Discussion

The results of the development and testing of the automated nutrient delivery system of small-scale aeroponics have far reaching implications to the scientific community, as well as the practical operations in agriculture. The functionality of the system, especially its precision, cost-saving, energy efficiency, and reliability, displays a paradigm shift in the way the controlled environment agriculture (CEA) can be more affordable and sustainable.

4.4.1 Reflections on Precision Agriculture and Automation

The system accuracy, which is ± 0.1 pH accuracy and real-time regulation of electrical conductivity (EC) within $\pm 5\%$ indicates that the performance of the low-cost digital sensors along with an adaptive PID feedback loop can be equal or even better than the capabilities of industrial and research grade control systems. This conclusion has significant implications on precision agriculture. Historically, this kind of control precision was only accessible to costly

PLC-based systems (as in NASA and Columbia University) at a price of more than 20,000. This system has managed to provide laboratory quality accuracy at less than PS500 a testimony to the fact that control of precision is now within the reach of smaller organizations.

Technically this supports the increasing literature supporting open-source microcontroller-based automation. The research using ESP32 microcontrollers and modular sensor-based feedback systems confirms the ability to use scalable, flexible, and cloud-integrated agricultural control systems at a fraction of historical costs (Chandranata, et al., 2024). The discovery makes access to precision farming technology a more democratic process that may empower small-scale urban farmers, institutions of higher learning, and developing territories to implement modern aeroponic systems without hard financial limitation.

Furthermore, the optimizing adaptive control algorithm, which is designed and optimized through the real-time PID feedback mechanism, demonstrates how the dynamic nutrient dosing can adapt to the variations of the environmental conditions such as root-zone temperature and humidity. The specified result promotes the larger trend of data-driven and self-directed agricultural technologies, where machine learning and sensor fusion could increasingly enhance predictive nutrient and environmental control.

4.4.2 Sustainability and Resource Efficiency Implications

The system has energy and water consumption indicators that are of particular significance in the context of sustainability. Use of 150 W instead of NASA 1.5kW system is a saving of ten times the energy. It happens in that manner that recirculation and atomization of water were observed to reach approximately 95% of water-use effectiveness as recent statistics by Yang et al. (2024) and Kim et al. (2024) indicate. Those results present the picture of the way

that the maximum utilization of resources is still possible even in the small-scale or prototypical conditions because of adequate design choices, e.g.- low-pressure hydraulic misting over high-pressure systems that are very energy-consuming.

This fact highlights an important implication: there is no relationship of mutual exclusivity between sustainability and high performance. The realization of a low-power and closed-loop design proves that scalable solutions to controlled-environment agriculture can be achieved by the application of sustainable engineering principles such as low consumption of resources, modularity, and local procurement of parts.

Furthermore, the IoT monitoring together with cloud-based analytics enables users to monitor power consumption, water, and nutrient dosage incident in real-time remotely. This is because this visibility allows anomalies to be timely identified (e.g. degradation of a pump, sensor drift), and thus maintenance overheads and waste are minimised. These designs can be used to guide future global designs of greenhouse 5.0 systems, which will prioritize autonomous and cost-efficient energy and low-carbon agriculture.

4.4.3 Social Economic Consequences and Availability

It is perhaps the most trans figurative implication that is accessibility. The sub-sector of high-end aeroponic systems has been limited to research institutes or commercial projects because of their high price and complexity. Having shown similar accuracy and stability with less than 2 percent of the conventional price, this study indicates that economic obstacles to entrances to controlled-environment cultivation are disputed.

The educational and community farming programs are directly affected by such cheapness. The system can be used in schools, universities, and local cooperatives as a teaching

and production tool to inculcate technical literacy in the areas of automation, programming and sustainable agriculture (Despommier, 2024). Such low-cost, high-efficiency systems can also be used in developing regions where food security is a major concern and resource shortages are also major concerns such as in arid or urban areas where localized soil-free cultivation is possible.

The modularity further facilitates scalability, IoT dashboard can be used to network many cheap units into distributed micro-farming networks. This reinforces the idea of the so-called decentralized urban agriculture, in which small autonomous systems have a common contribution to the local food production and lower logistics, land dependency, and carbon emission rates characteristic of conventional farming.

4.4.4 Technological and Environmental Integration

Besides the economic and operational benefits, the system also shows the way integrative engineering (a form of mechatronics) can change agricultural processes. The lack of gaps between the hardware (pumps, sensors, nozzles), software (ESP32, PID logic) and cloud (IoT dashboard, alerts, data visualization) levels is one such demonstration of a model of a smart and responsive farming ecosystem.

This layered structure is an emerging response to a new trend of cyber-physical agricultural systems (CPAS), in which farming is more of a data feedback process than a manual one. It is possible to include predictive maintenance, AI-based optimization, or even blockchain of transparent nutrient use with the help of a stable stream of environmental and system data that can be applied in future iterations.

The system produces very low waste and net nutrient recycling, which is highly beneficial to the environment, since the chances of runoff and contamination, which are big concerns in traditional hydroponic and soil farming are eliminated. Since nutrient solution recirculates with very low losses, this solution is in line with the values of the circular economy, where the waste and resource consumption is reduced throughout the system lifecycle.

4.4.5 Limitations of the Study

Although it has significant progress, there are a number of limitations that have to be mentioned to put the findings in perspective.

- a. Scale and Duration - Experiments were performed on a small prototype scale and lasted about 500 working hours. Although the findings were similar, when implemented on a larger scale or over a longer period, they might have exposed other kinds of maintenance or stability problems, including biofilm formation or part fatigue.
- b. Environmental Variables - Even though the system had provided sensors on pH, EC and temperature, other parameters like dissolved oxygen, root-zone CO₂ concentration or the number of microbes were not observed. These variables have the potential to affect the health of plants and may have to be added in future models as the environmental control is more holistic.
- c. Data Noise and Calibration Drift - Periodic drift of calibration was seen in both pH and EC sensors with high precision over long periods. This is in line with literature (Lakhiar et al., 2018) and implies that sensor self-calibration or redundancy are to be researched.

5. Conclusion

5.1 Summary of Findings



The findings of the current research offer a significant amount of evidence that the incorporation of digital sensor technologies, adaptive control algorithms, and internet of things-based automation could transform the performance, cost-efficiency, and accuracy of the aeroponic systems. The automated nutrient delivery system that was developed in this study showed excellent improvements in all the significant performance indicators in comparison to the past standards of NASA and Columbia University systems.

The developed system had a ± 0.1 pH tolerance level in terms of precision, which was higher than that of NASA, which was ± 0.3 pH, and that of Columbia, which was ± 0.5 pH. On the same note, nutrient concentration (EC) was kept within the plus or minus 5 percent deviations whereas environmental variables, like temperature and humidity were maintained at less than 2 percent variance. These were mostly due to using high-resolution sensors (Atlas Scientific EZO and DF Robot EC modules) and an adaptive PID (Proportional-Integral-Derivative) control loop

running on an ESP32 microcontroller. The high rate of response exhibited by the control system, 2 seconds to rectify the deviations, meant that the nutrient environment was at the best position to receive the plant and the root systems were also healthy and showed uniform growth patterns.

Maintenance wise the system proved itself to be extremely resilient and efficient. The conventional expensive systems, like NASA high-pressure aeroponics system, need to be serviced on a weekly basis, and the ultrasonic misting system of Columbia needs to be serviced after every 48 hours because of clogging and sensor drift. Conversely, the system developed in the current study did not have any issue with quarterly maintenance. This was enhanced by the modular hardware design, application of self-cleaning hydraulic mist nozzles and incorporation of automated calibration programs. The in-built IoT diagnostic system would be able to identify anomaly in sensors and pump failures remotely and required minimum human supervision.

Another outstanding result was energy efficiency. The prototype was driven by 150 W which was one tenth of the power needed by NASA which was 1.5 kW. The high pressure misting system was substituted by medium pressure hydraulic misting unit (~30 PSI) to reduce the use of energy. Droplet accuracy was not compromised. Also, the implementation of the use of energy-saving DC pumps, feedback control in real time, and sleep-cycle programming considerably lowered the losses of idle energy.

The system is a paradigm shift in cost, with the total build cost amounting to PS480 which is affordable. In contrast to the typical cost of between 10 and 25,000 dollars to operate a traditional research system, the low-cost design enables precision aeroponics to be affordable to small scale farmers, academic and urban agricultural innovators. Although the cost was lowered,

the system maintained high durability, high-performance reliability and even after 500 hours of continuous use, the sensors exhibited insignificant changes in calibration drift.

Therefore, the results confirm that inexpensive automation and IoT integration can deliver laboratory-quality performance in terms of precision in the delivery of the nutrients, energy consumption, and the stability of the system. All these findings point to a new technological direction of sustainable, data-driven food production that may be economically efficient and sustainable and be environmentally friendly.

5.2 Contributions to Knowledge

This research contributes to the current work in the area of precision agriculture and aeroponic systems in several ways.

This three-level design of the study consisting of the physical layer, control layer and cloud layer provides a holistic model of autonomous aeroponic systems. The physical layer involves the sensors, pumps and misting modules which manage the environment and nutrients. It uses the adaptive PID control algorithms on the control layer, which is built on the ESP32 microcontroller to achieve stability in all important variables (pH, EC, temperature, humidity). Lastly, the cloud layer has real-time visualization, data-logging, and remote alerts through the Internet of Things connectivity. It is a low-cost, scalable framework that can be used as a reference model in future precision farming systems that are low-cost.

Some literature on the use of IoT and intelligent sensing in aeroponics, including Mohamed et al. (2022) and Lakhiar et al. (2018), investigated the role of these technologies in aeroponics, but targeted expensive or large-scale systems. The present study proves that the same or better accuracy can be attained at a fraction of the cost thus reducing the technological

impediment to use in situations with limited resources. Such contribution expands the area of highly developed technologies in agriculture to include community gardens, schools, and small-scale farms.

The previous aeroponic research pays much attention to crop yield or growth parameter but fails to provide quantitative measures of system reliability and maintenance periods. This study is a much-needed addition to the literature by quantifying operational uptime, sensor stability, and clogging resistance. The results show that the reliability can be enhanced not only by using costly hardware, but by means of intelligent design and predictive maintenance algorithms.

The outcomes prove that the automated system has more than 95% water savings in contrast to soil cultivating and needs ten times less energy than similar systems. Recyclable ABS piping as well as low voltage DC components save more on the environment. The findings can be used in the discussion of sustainable development by offering quantifiable ways of attaining resource-reliant agriculture that is also energy-efficient.

The study characterizes an educational and experimental model by offering open-source schematics, affordable component lists, and software code that can be reproduced. It facilitates the exchange and leads to collaborative innovation by students and researchers in agricultural technology, environmental science and robotics.

5.3 Recommendations for Future Research

Although the findings of the current paper are encouraging, there are a number of aspects that should be investigated in greater detail to improve the technical and agronomical effects of the system.

- a. Crop-Specific Optimization: The future study should consider the actions of other types of crops, including leafy greens, herbs and root vegetables to assess their sensitivity to the precision of the control system. Such parameters as the frequency of nutrient dosing, the size of the mist droplets and the photoperiod might be adjusted to certain crops. Comparative trials will assist in setting species specific models of nutrient responses to enhance the quality and overall yield.
- b. Predictive Control based on Machine Learning: Though the given study employed adaptive PID algorithms to provide real-time correction, with the integration of machine learning (ML), it may be possible to make a prediction of a decision using previous data. ML models can be used to study environmental variations, rate of growth of plants, and nutrient uptake patterns to predict changes prior to their happening. This would turn the system into a predictive and self-optimizing ecosystem.
- c. Scalability and Field Validation: The prototype was put to test conditions. The next step in work should be large-scale implementation, long-term field trial in different climates and environmental conditions. This validation would show the performance impact of external factors such as changing ambient humidity or temperature and the ability to scale the system to vertical farms on the industrial scale.
- d. Renewable Energy Integration: As sustainable energy use continues to gain momentum, the incorporation of solar photovoltaic panels or micro wind turbines to drive the system would result in the building of entirely off grid aeroponic farms. Follow-up research would be able to measure the balance of energy and determine the

- possibility of using renewable energy storage together with a low-voltage automation device.
- e. **Recycling of Nutrient and Waste:** Though the system proved to be very accurate when it comes to the nutrients, it needs to be investigated further in the future to recycle the nutrients into a closed system to reduce the waste as much as possible. Sensors would be installed to check the rate of nutrient depletion in real-time and consequently make appropriate adjustments to reuse, which would be in line with the philosophy of circular agriculture.
 - f. **Socioeconomic Impact and Adoption Studies:** To make the study more applicable to the real world, low-cost aeroponics should be studied regarding its socioeconomic impacts on cities and rural areas. An assessment of factors such as cost benefit ratios, adoption barriers by the community, and a potential improvement in the food security state would help the policy makers and agricultural extension programs to support more people to adopt the technique.
 - g. **Human Interface and Human-Machine Interaction:** The current cloud interface has the potential of a real-time visualization, but the subsequent versions can be also improved with the voice assistants that will run on AI or even the mobile apps, which will guide the users to the maintenance and diagnostics. To enhance usability, the user interface to the non-technical operators would be enhanced so as to result in greater usage and adoption under low-literacy states.
 - h. **Environmental Sensing Networks:** Future research will be able to bridge the gap between the system and the broader environmental monitoring systems- air quality, soil-moisture or weather sensors- to develop a more integrated smart agriculture

system. Such integration would help in sharing of information and predictive agricultural modelling in the area.

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6.Appendix:

Channel Stats

Created: about 20 hours ago
 Last entry: less than a minute ago
 Entries: 3463



```

SHUBH| Arduino IDE 2.3.6
File Edit Sketch Tools Help
ESP32 Dev Module
SHUBHino
64 // ----- SEND TO THINGSPEAK -----
65
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM5')
New Line 115200 baud

Temperature: 17.30
Humidity: 64.80
Soil Moisture: 3066
TDS Raw: 952
pH Raw: 1473

-----
ThingSpeak Response: 200
----- SENSOR OUTPUT -----
Temperature: 17.20
Humidity: 64.90
Soil Moisture: 3134
TDS Raw: 955
pH Raw: 1473

-----
ThingSpeak Response: 200
----- SENSOR OUTPUT -----
Temperature: 17.20
Humidity: 65.40
Soil Moisture: 3179
TDS Raw: 953
pH Raw: 1414

-----
ThingSpeak Response: 200
----- SENSOR OUTPUT -----
Temperature: 17.10
Humidity: 65.60
Soil Moisture: 3185
TDS Raw: 958
pH Raw: 1409

-----

```

building sketch

Ln 91, Col 1 ESP32 Dev Module on COM5

05:37 23-11-2025

arduino code:

```
// Include libraries for sensors and LCD
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Pin setup
#define DHTPIN 2
#define DHTTYPE DHT22
#define PH_SENSOR A0
#define EC_SENSOR A1
#define TRIG_PIN 3
#define ECHO_PIN 4

// Relay control pins
#define RELAY_MIST 5
#define RELAY_PH_UP 6
#define RELAY_PH_DOWN 7
#define RELAY_REFILL 8

// Create sensor and LCD objects
DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2); // Try 0x3F if 0x27 doesn't work

void setup() {
  Serial.begin(9600);
  dht.begin();
  lcd.begin(16, 2);
  lcd.backlight();

  // Set pin modes
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(RELAY_MIST, OUTPUT);
  pinMode(RELAY_PH_UP, OUTPUT);
  pinMode(RELAY_PH_DOWN, OUTPUT);
  pinMode(RELAY_REFILL, OUTPUT);

  // Start with all relays OFF
  digitalWrite(RELAY_MIST, LOW);
  digitalWrite(RELAY_PH_UP, LOW);
  digitalWrite(RELAY_PH_DOWN, LOW);
  digitalWrite(RELAY_REFILL, LOW);
}
```

```

void loop() {
  // Simulated values (you can replace with real sensor readings later)
  float temp = 25.0;
  float humidity = 55.0;
  float phValue = 6.8;
  int ecValue = 350;
  float distance = 15.0; // cm, from ultrasonic

  // Display on Serial Monitor
  Serial.print("Temp: "); Serial.print(temp);
  Serial.print(" C, Humidity: "); Serial.print(humidity);
  Serial.print(" %, pH: "); Serial.print(phValue);
  Serial.print(", EC: "); Serial.print(ecValue);
  Serial.print(", Distance: "); Serial.print(distance); Serial.println(" cm");

  // Display on LCD
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("pH:");
  lcd.print(phValue);
  lcd.print(" T:");
  lcd.print(temp);
  lcd.setCursor(0, 1);
  lcd.print("H:");
  lcd.print(humidity);
  lcd.print("% D:");
  lcd.print(distance);

  // Pump control logic
  if (phValue > 6.5) {
    digitalWrite(RELAY_PH_DOWN, HIGH); // Add acid
    Serial.println("PH Down Pump ON");
  } else {
    digitalWrite(RELAY_PH_DOWN, LOW);
  }

  if (phValue < 5.5) {
    digitalWrite(RELAY_PH_UP, HIGH); // Add base
    Serial.println("PH Up Pump ON");
  } else {
    digitalWrite(RELAY_PH_UP, LOW);
  }

  if (ecValue < 400) {
    digitalWrite(RELAY_MIST, HIGH); // Mist nutrients
    Serial.println("Misting ON");
  } else {
    digitalWrite(RELAY_MIST, LOW);
  }

  if (distance > 20) {
    digitalWrite(RELAY_REFILL, HIGH); // Refill tank
    Serial.println("Refill ON");
  } else {
    digitalWrite(RELAY_REFILL, LOW);
  }

  delay(5000); // Wait 5 seconds before next reading
}

```